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# Experiment 07

## VLAN Configuration, Router-on-a-Stick & Multi-VLAN DHCP Server

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### Aim

To implement a router as a DHCP server that can serve multiple VLANs, using Cisco Packet Tracer.

### Learning Objectives

After completing this experiment, we should be able to:

1. Understand the concept of VLANs.
2. Create and configure multiple VLANs on a switch.
3. Assign switch ports to appropriate VLANs.
4. Understand the difference between an Access Port and a Trunk Port.
5. Configure a trunk between a switch and a router.
6. Understand Inter-VLAN Routing and the Router-on-a-Stick technique.
7. Configure router subinterfaces for multiple VLANs.
8. Configure a router as a DHCP Server with separate DHCP pools per VLAN.
9. Verify VLAN, trunk, DHCP, and routing configurations.

### Theory

#### VLANs and Broadcast Domains

A **VLAN** (Virtual Local Area Network) is a logical network created within a physical switch. Instead of placing every connected device into the same broadcast domain, a switch's ports can be logically grouped into separate VLANs (for example, one VLAN for students and another for faculty), even though every device shares the same physical switch. This gives better network segmentation, improved security, a smaller broadcast domain per group, and easier management.

#### Inter-VLAN Routing and Router-on-a-Stick

Each VLAN is a separate Layer 2 network, so a switch alone cannot forward traffic between two different VLANs – a Layer 3 device (a router or a Layer 3 switch) is required. This process is called **Inter-VLAN Routing**. **Router-on-a-Stick** performs inter-VLAN routing using a single physical router interface that is divided into multiple logical **subinterfaces** – one per VLAN – so that one physical port can serve every VLAN instead of needing a dedicated cable per VLAN. Each subinterface is assigned an IP address that acts as the default gateway for its VLAN.

#### VLAN Tagging (IEEE 802.1Q)

When several VLANs share a single trunk link, the receiving device needs to know which VLAN each frame belongs to. This is achieved with **802.1Q VLAN tagging**, which inserts a VLAN identifier into the Ethernet frame as it crosses the trunk. On the router, the command

`encapsulation dot1Q <vlan-id>` tells a subinterface which VLAN's tagged frames it should handle.

## DHCP and the DORA Process

**DHCP** (Dynamic Host Configuration Protocol) automatically assigns network configuration – IP address, subnet mask, default gateway, and DNS server – to client devices, removing the need to configure every host manually. A Cisco router can itself act as a DHCP server. The DHCP exchange follows four steps, known as **DORA**:

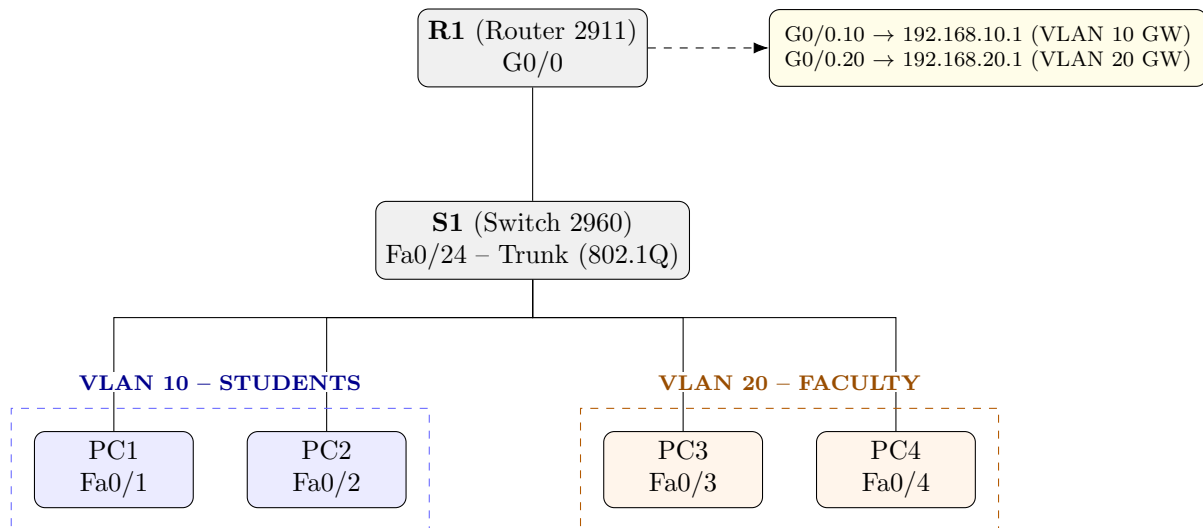
- **D – Discover:** the client broadcasts a request asking if a DHCP server is available.
- **O – Offer:** the server offers an available IP address.
- **R – Request:** the client requests the offered address.
- **A – Acknowledgment:** the server confirms the allocation.

When one router serves multiple VLANs, it maintains a **separate DHCP pool per VLAN**, each pool tied to that VLAN's subnet, so that VLAN 10 clients receive addresses from the VLAN 10 pool and VLAN 20 clients receive addresses from the VLAN 20 pool. The address already used by the router's own subinterface (the gateway) is marked with `ip dhcp excluded-address` so that DHCP never hands it out to a client.



## Network Topology

The router (R1) connects to the switch (S1) over a single trunk link carrying both VLANs; the switch's remaining ports are split between the two VLANs, two access ports each.



### Devices Used

1 Router (2911) – R1; 1 Switch (2960) – S1; 4 PCs – PC1, PC2, PC3, PC4. All PC-to-switch and switch-to-router links use Copper Straight-Through cables.

### VLAN and Addressing Plan

| VLAN | Name     | Network         | Gateway      |
|------|----------|-----------------|--------------|
| 10   | STUDENTS | 192.168.10.0/24 | 192.168.10.1 |
| 20   | FACULTY  | 192.168.20.0/24 | 192.168.20.1 |

### Device / Port / VLAN Mapping

| Device | Switch Port | VLAN           |
|--------|-------------|----------------|
| PC1    | Fa0/1       | VLAN 10        |
| PC2    | Fa0/2       | VLAN 10        |
| PC3    | Fa0/3       | VLAN 20        |
| PC4    | Fa0/4       | VLAN 20        |
| Router | Fa0/24      | Trunk (802.1Q) |



## Procedure

### Part A – Configure the Switch (S1)

#### *Step 1 – Enter privileged and global configuration mode*

```
Switch> enable
Switch# configure terminal
Switch(config)#
```

#### *Step 2 – Create VLAN 10 and VLAN 20*

```
Switch(config)# vlan 10
Switch(config-vlan)# name STUDENTS
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name FACULTY
Switch(config-vlan)# exit
```

#### *Step 3 – Verify that the VLANs were created*

```
Switch(config)# do show vlan brief
```

#### *Step 4 – Assign PC1 and PC2 (Fa0/1–Fa0/2) to VLAN 10*

```
Switch(config)# interface range fastEthernet 0/1-2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

#### *Step 5 – Assign PC3 and PC4 (Fa0/3–Fa0/4) to VLAN 20*

```
Switch(config)# interface range fastEthernet 0/3-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

#### *Step 6 – Configure Fa0/24 as a trunk port (link to the router)*

```
Switch(config)# interface fastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```

#### *Step 7 – Verify the trunk*

```
Switch# show interfaces trunk
```

Fa0/24 should now be listed as a trunk port, confirming the switch is ready to carry both VLANs toward the router.



## Part B – Configure the Router (R1)

### *Step 1 – Enter privileged and global configuration mode*

```
Router> enable
Router# configure terminal
Router(config)#
```

### *Step 2 – Bring up the physical interface G0/0*

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit
```

### *Step 3 – Create the subinterface for VLAN 10*

```
Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit
```

### *Step 4 – Create the subinterface for VLAN 20*

```
Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

### *Step 5 – Verify the router interfaces*

```
Router# show ip interface brief
```

Both G0/0.10 and G0/0.20 should show as up/up, with 192.168.10.1 and 192.168.20.1 respectively. This is **Router-on-a-Stick**: one physical interface, two logical subinterfaces, each acting as the gateway for its own VLAN, connected to the switch over a single trunk link.

### *Step 6 – Exclude the gateway addresses from DHCP*

```
Router(config)# ip dhcp excluded-address 192.168.10.1
Router(config)# ip dhcp excluded-address 192.168.20.1
```

### *Step 7 – Create the DHCP pool for VLAN 10*

```
Router(config)# ip dhcp pool VLAN10_POOL
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# exit
```

### *Step 8 – Create the DHCP pool for VLAN 20*

```
Router(config)# ip dhcp pool VLAN20_POOL
```



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```
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# exit
```

### Part C – Configure the PCs

On each PC, go to **Desktop** → **IP Configuration** and select **DHCP** instead of entering an address manually. PC1 and PC2 (VLAN 10) should automatically receive addresses in 192.168.10.0/24; PC3 and PC4 (VLAN 20) should receive addresses in 192.168.20.0/24.



## Verification / Output

*show vlan brief (on S1)*

| VLAN | Name     | Status | Ports        |
|------|----------|--------|--------------|
| 1    | default  | active |              |
| 10   | STUDENTS | active | Fa0/1, Fa0/2 |
| 20   | FACULTY  | active | Fa0/3, Fa0/4 |

*show interfaces trunk (on S1)*

| Port   | Mode                   | Encapsulation | Status   | Native vlan |
|--------|------------------------|---------------|----------|-------------|
| Fa0/24 | on                     | 802.1q        | trunking | 1           |
| Port   | Vlans allowed on trunk |               |          |             |
| Fa0/24 | 1,10,20                |               |          |             |

*show ip interface brief (on R1)*

| Interface             | IP-Address   | Status | Protocol |
|-----------------------|--------------|--------|----------|
| GigabitEthernet0/0    | unassigned   | up     | up       |
| GigabitEthernet0/0.10 | 192.168.10.1 | up     | up       |
| GigabitEthernet0/0.20 | 192.168.20.1 | up     | up       |

*IP Configuration received by each PC (via DHCP)*

| PC  | IP Address   | Subnet Mask   | Default Gateway |
|-----|--------------|---------------|-----------------|
| PC1 | 192.168.10.2 | 255.255.255.0 | 192.168.10.1    |
| PC2 | 192.168.10.3 | 255.255.255.0 | 192.168.10.1    |
| PC3 | 192.168.20.2 | 255.255.255.0 | 192.168.20.1    |
| PC4 | 192.168.20.3 | 255.255.255.0 | 192.168.20.1    |

*Note: the exact addresses assigned can vary slightly depending on lease order in Packet Tracer; the important part is that each PC lands in the correct VLAN's subnet with the correct gateway.*

*show ip dhcp binding (on R1)*

| IP address   | Client-ID/Hardware address | Lease expiration |
|--------------|----------------------------|------------------|
| 192.168.10.2 | 0100.5079.6668.0a          | ...              |
| 192.168.10.3 | 0100.5079.6668.0b          | ...              |
| 192.168.20.2 | 0100.5079.6668.0c          | ...              |
| 192.168.20.3 | 0100.5079.6668.0d          | ...              |

## Result

The switch was configured with two VLANs (STUDENTS and FACULTY) and a trunk link to the router. Using Router-on-a-Stick, the router's single physical interface was split into two subinterfaces, one per VLAN, each acting as that VLAN's default gateway. The router was then configured as a DHCP server with a separate address pool per VLAN, so that PC1/PC2 automatically obtained addresses in 192.168.10.0/24 and PC3/PC4 automatically obtained addresses in 192.168.20.0/24 – confirming that a single router can simultaneously provide inter-VLAN routing and DHCP services to multiple VLANs.



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## Learning Outcome

After completing this experiment, the following learning outcomes were achieved.

- Understood how VLANs logically segment a single physical switch into separate broadcast domains.
- Learned to create VLANs and assign access ports to them using `switchport access vlan`.
- Understood the difference between an access port (carries one VLAN, untagged) and a trunk port (carries multiple VLANs, 802.1Q-tagged).
- Implemented Router-on-a-Stick using subinterfaces and `encapsulation dot1q` to perform inter-VLAN routing over a single physical link.
- Configured a Cisco router as a DHCP server with per-VLAN address pools, excluded gateway addresses, and default-router/DNS options.
- Verified the configuration end-to-end using `show vlan brief`, `show interfaces trunk`, `show ip interface brief`, and `show ip dhcp binding`.
- Gained practical, hands-on exposure to building and testing a multi-VLAN, DHCP-enabled network in Cisco Packet Tracer.